

Penetration Testing – Task 1 Report

Week 5: Reconnaissance & Information Gathering

1. Objective

The objective of this task was to perform reconnaissance and information gathering on the target system (**Metasploitable2**) in a controlled lab environment, to identify the network architecture, open ports, services, operating system details, and potential attack vectors.

2. Target Information

- **Target Machine:** Metasploitable2
 - **Target IP:** 192.168.0.102
 - **Environment:** VirtualBox
 - **Attacker Machine:** Kali Linux
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3. Tools Used

- whois – IP and ownership details lookup
 - nmap – Network scanning and service enumeration
 - traceroute – Network path mapping
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4. Methodology & Execution

Step 1 – WHOIS Lookup

Command:

```
whois 192.168.0.102
```

Result:

- IP belongs to the **private address range** (RFC1918 – 192.168.0.0/16)
 - Allocated for local network use; no public registry ownership details.
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Step 2 – Network Path Tracing

Command:

```
tracert 192.168.0.102
```

Result:

- Only **1 hop** to the target, confirming both machines are on the same local network.

Step 3 – Network & Service Scan

Command:

```
nmap -A -T4 192.168.0.102
```

Key findings:

- **Host Status:** Up (0.0012s latency)
- **OS Detected:** Linux Kernel 2.6.9 – 2.6.33
- **MAC Address:** VirtualBox Adapter

Open Ports & Services:

Port	Service	Version / Details
21	FTP	vsftpd 2.3.4 (Anonymous login enabled)
22	SSH	OpenSSH 4.7p1 Debian
23	Telnet	Linux telnetd
25	SMTP	Postfix smtpd (SSLv2 enabled)
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache 2.2.8 (Ubuntu)
111	RPC	rpcbind + NFS services
139	NetBIOS	Samba smbd 3.X
445	SMB	Samba smbd 3.0.20
512	Exec	netkit-rsh rexecd
513	Login	rlogin service

Port	Service	Version / Details
1099	Java RMI	GNU Classpath grmiregistry
1524	Bind Shell	Root shell access
2049	NFS	Versions 2-4 supported
2121	FTP	ProFTPD 1.3.1
3306	MySQL	v5.0.51a
5432	PostgreSQL	v8.3.x
5900	VNC	Protocol 3.3
6000	X11	Access denied
6667	IRC	UnrealIRCd
8009	AJP13	Apache JServ Protocol
8180	HTTP	Apache Tomcat 5.5

5. Analysis of Potential Attack Vectors

From the reconnaissance:

1. **FTP Anonymous Login** – Potential file disclosure or upload exploit.
2. **Outdated Services** – vsftpd 2.3.4, OpenSSH 4.7p1, and Apache 2.2.8 have known vulnerabilities.
3. **Weak Security Protocols** – SSLv2 enabled on SMTP (deprecated & insecure).
4. **Multiple Remote Access Services** – Telnet, SSH, Rlogin, and VNC all increase attack surface.
5. **Exposed Databases** – MySQL and PostgreSQL accessible from network.
6. **Misconfigured SMB** – Message signing disabled (possible relay attacks).
7. **Default/Old Web Applications** – Apache Tomcat 5.5 with potential default credentials.

6. Conclusion

The target (Metasploitable2) exhibits multiple high-risk vulnerabilities based on open services and outdated software. These findings will guide the next phase: **Vulnerability Scanning** using Nessus or OpenVAS.